LESSON 1.5

Math is Finding Patterns

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

WHY THIS MATTERS Akela's change jar is full of pennies, nickels, dimes, and quarters. The value of the coins is $53.50. How many of each coin could be in the jar? There is more than one answer — and patterns are the way in.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
pattern *Spanish: patrón*	Something that repeats or changes in a predictable way.	1, 3, 7, 15, 31 — each tower's steps follow a predictable jump.
pattern rule *Spanish: regla del patrón*	The instruction that tells how to get from one number in a pattern to the next.	2, 4, 6, 8 … the rule is "+ 2 every time".
generalization *Spanish: generalización*	A statement that is true for a whole pattern, made by noticing repeated calculations.	1 → 3, 2 → 5, 3 → 7 … so n → 2n + 1, for ANY n.
table of values *Spanish: tabla de valores*	A table that lists each step of a pattern so you can check that a solution is reasonable.	discs 1, 2, 3 beside steps 1, 3, 7 — one row per step.
reasonableness *Spanish: sensatez del resultado*	Whether an answer makes sense — checked as you work, with adjustments made when needed.	5/6 of 540 should land BELOW 540, so 650 fails the check.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A change jar is full of pennies, nickels, dimes, and quarters worth $53.50 in total. How many of each coin could there be?

1. Each penny has a value of $0.01, so 100 pennies have a value of $1.00.
2. Each nickel has a value of $0.05, so 100 nickels have a value of $5.00. Each dime has a value of $0.10, so 100 dimes have a value of $10. Each quarter has a value of $0.25, so 100 quarters have a value of $25.
3. I multiply the number of each coin by its value. And I notice a generalization: because the jar's total, $53.50, ends in 0, the number of pennies can only be a multiple of 5.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Try 200 quarters: 200 × $0.25 = $50.00.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ is only a one-time fact because _____; it would be a pattern if _____.
- I noticed that _____ .
- This helps because instead of counting one at a time, I can _____ .
- I look for _____ .

Word bank pattern | pattern rule | generalization | table of values | reasonableness

WATCH OUT!

A common mistake

Reporting only the points from the streak (20 or 30) and forgetting to add the points each player had before the streak began.

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Each penny has a value of $0.01. What is the value of 100 pennies?

(A) $0.10 (B) $1.00

(C) $10.00 (D) $100.00

STEPS TO TRY

1. A penny is one hundredth of a dollar.
2. How many hundredths make a whole?
3. 100 × 0.01 = ?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Each quarter has a value of $0.25. Following the pattern, what is the value of 100 quarters?

(A) $2.50
(B) $25.00
(C) $100.25
(D) $250.00

STEPS TO TRY

1. Multiplying by 100 moves the decimal point.
2. 100 pennies → $1.00, 100 dimes → $10. Same move.
3. 100 × 0.25 = ?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Player A scores a two-pointer on every shot. What is Player A's pattern rule?

(A) Add 2
(B) Multiply by 2
(C) Add 3
(D) Add 18

STEPS TO TRY

1. How much is one two-pointer worth?
2. Each made shot changes the score by the same amount.
3. The score goes up by 2 each time.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Player B had 6 points, then made 10 three-pointers in a row. How many points does Player B have now?

(A) 9
(B) 16
(C) 30
(D) 36

STEPS TO TRY

1. How many points do 10 three-pointers add?
2. 10 × 3 = 30.
3. Now add the starting 6.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

SENTENCE STARTER ___ is only a one-time fact because ___; it would be a pattern if ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help